

Lab Assignment 7

1 Experiment 1: Limitations of Logistic Regression

1.1 Objective

The purpose of this experiment is to explore situations in which Logistic Regression may fail to produce reliable classification results. Although Logistic Regression works well for linearly separable data, its performance can degrade when the underlying structure of the data is more complex.

1.2 Tasks

1. Generate three synthetic datasets with different characteristics:
 - **Dataset A:** A dataset in which the two classes are clearly linearly separable.
 - **Dataset B:** A dataset where the classes overlap significantly.
 - **Dataset C:** A dataset where the class boundary follows a circular or non-linear pattern.
2. Train a Logistic Regression classifier separately on each dataset.
3. For every dataset:
 - Visualize the distribution of data points.
 - Plot the decision boundary learned by the classifier.
 - Compute the classification accuracy.
 - Generate the confusion matrix.
4. Compare the model behaviour across all three datasets.

1.3 Discussion

Provide explanations for the following observations:

- Why Logistic Regression performs well when the data is linearly separable (Dataset A).
- Why the model performance decreases when the classes overlap (Dataset B).
- Why Logistic Regression struggles to correctly model the circular decision boundary in Dataset C.

2 Experiment 2: Impact of Irrelevant Features

2.1 Objective

This experiment investigates how the presence of irrelevant or noisy features influences the learning behaviour of classification models.

2.2 Dataset Construction

Begin with a dataset that contains two informative features responsible for determining the class label. Then gradually introduce additional random features that do not contribute to the prediction task.

The experiments should follow the configuration below:

Experiment	Feature Composition
Exp 1	2 informative features
Exp 2	2 informative + 5 random features
Exp 3	2 informative + 20 random features
Exp 4	2 informative + 50 random features

2.3 Tasks

For each experimental setup:

1. Train the following classifiers:

- Logistic Regression
- Gaussian Naive Bayes

2. Evaluate both models using:

- Classification accuracy
- Training time
- Confusion matrix

2.4 Discussion

Discuss the following points:

- Whether the introduction of irrelevant features negatively affects model performance.
- Which of the two classifiers appears more robust in the presence of noisy or redundant features.

3 Experiment 3: Decision Boundary Analysis

3.1 Objective

The aim of this experiment is to analyze how the separability of a dataset influences the performance of Logistic Regression when trained using Gradient Descent.

3.2 Dataset Generation

Generate two datasets, each containing 500 samples.

Dataset A: Linearly Separable

$$z = 2x_1 + 3x_2 + \epsilon$$

Dataset B: Non-linear Boundary

$$z = x_1^2 + x_2^2 - 1 + \epsilon$$

where

$$x_1, x_2 \sim U(-2, 2)$$

$$\epsilon \sim N(0, 0.2)$$

The class labels are defined as

$$y = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}$$

3.3 Tasks

1. Train a Logistic Regression model using Gradient Descent.
2. During training, record the following values at every epoch:
 - Training accuracy

- Testing accuracy
 - Loss value
3. Generate the following plots:
 - Accuracy vs Epoch
 - Loss vs Epoch
 4. Plot the learned decision boundary for both datasets.

3.4 Analysis

Discuss the following observations:

- Why Logistic Regression performs effectively for Dataset A.
- Why the model struggles to achieve high performance for Dataset B.
- Suggest alternative models that could better capture the non-linear boundary in Dataset B.